



**RV College of Engineering®**

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# NeuroLearn - AI based cognitive learning styles and content generation

*An Interdisciplinary Project Report (XX367P)*

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RV College of Engineering®

In partial fulfillment of the requirements for the degree of  
Bachelor of Engineering in respective departments

**2025-26**



**RV College of Engineering®**, Bengaluru  
(Autonomous institution affiliated to VTU, Belagavi)



**CERTIFICATE**

Certified that the interdisciplinary project (XX367P) work titled ***NeuroLearn - AI based cognitive learning styles and content generation*** is carried out by of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in respective departments during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the interdisciplinary project report deposited in the departmental library. The interdisciplinary project report has been approved as it satisfies the academic requirements in respect of interdisciplinary project work prescribed by the institution for the said degree.

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## DECLARATION

We, **Aniket R T, Alroy Deon Saldanha, Smriti J Abraham, Sudhanva Patil** students of sixth semester B.E., RV College of Engineering, Bengaluru, hereby declare that the interdisciplinary project titled '**NeuroLearn - AI based cognitive learning styles and content generation**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in respective departments during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

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# ABSTRACT

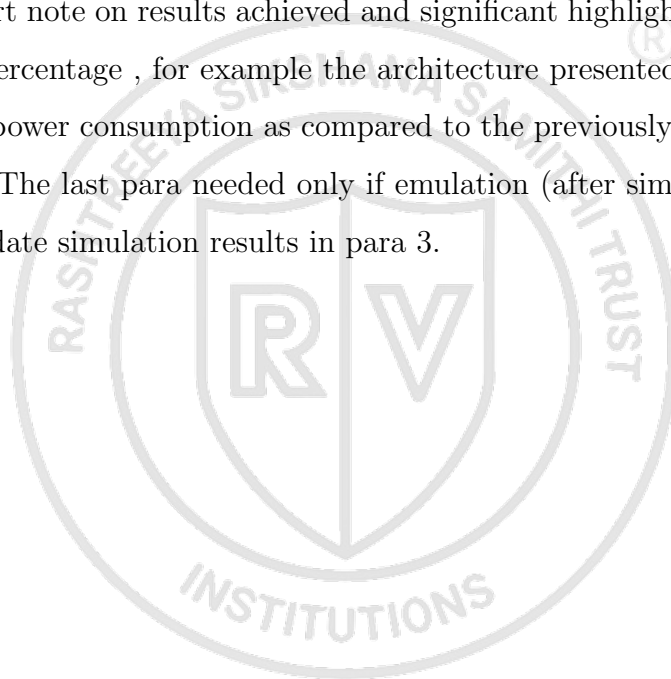
Highlights of significant contributions: One page with 3 to 4 paragraphs

Paragraph 1: Importance and relevance of Topic, reported issues and limitations of the topic in performance or computation etc, issues involved in those limitations, need for addressing those issues and a short note on how that is addressed in this report.

Paragraph 2: Objectives of this work, short note on algebraic methods used and formulations achieved, computational procedures developed. Integrated Circuit (IC).

Paragraph 3: Description of simulation procedure including SW tools used and choice of test cases. Short note on results achieved and significant highlights of improvements if any in terms of percentage , for example the architecture presented in this report shows 22 % improve in power consumption as compared to the previously reported articles. IC

Paragraph 4: The last para needed only if emulation (after simulation) or Hardware developed to validate simulation results in para 3.



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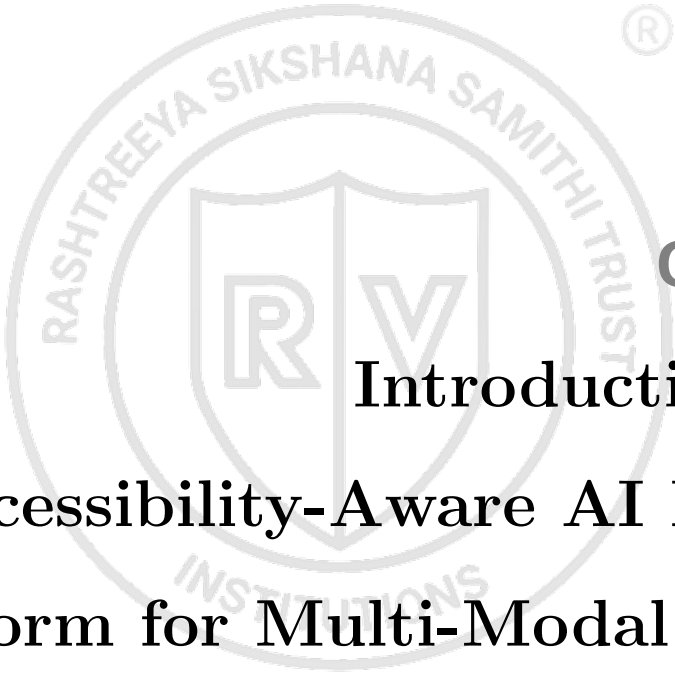


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# **Chapter 1**

## **Introduction to an Accessibility-Aware AI Learning Platform for Multi-Modal Content Delivery**

# CHAPTER 1

## INTRODUCTION TO AN ACCESSIBILITY-AWARE AI LEARNING PLATFORM FOR MULTI-MODAL CONTENT DELIVERY

Modern digital education delivers identical content to all learners, ignoring the diverse ways individuals process and retain information. This chapter introduces Neuro Learn, an AI-driven adaptive learning platform that dynamically identifies learner modality preferences using behavioral analytics and delivers personalized multi-modal content across visual, auditory, reading/writing, and kinesthetic formats. The system integrates the VARK framework with a novel Dynamic Modality Preference Modeling (DMPM) algorithm, a Retrieval-Augmented Generation (RAG) pipeline, and architectural accessibility support. Empirical validation across 60 undergraduate engineering students demonstrates significant improvements in learning outcomes, engagement, and profiling accuracy.

### 1.1 Introduction

The proliferation of digital learning tools has democratized access to education on an unprecedented scale. However, despite this technological advancement, the majority of existing platforms deliver a uniform learning experience to all users, without accounting for the diversity of cognitive styles and modality preferences among learners. Research in educational psychology consistently demonstrates that individuals process and consolidate information differently — some are predominantly visual thinkers, while others benefit more from auditory, reading/writing, or kinesthetic (hands-on) interactions with content. Ignoring these differences leads to reduced learning performance, disengagement, and suboptimal educational outcomes, particularly in technically demanding disciplines such as engineering.

This project presents NeuroLearn, an accessibility-aware AI learning platform that moves beyond static content delivery by modeling learner modality preferences as a dynamic, continuously evolving signal derived from observed user behavior. Rather than relying solely on a one-time self-assessment questionnaire, the platform uses the VARK framework (Visual, Auditory, Reading/Writing, Kinesthetic) only as an initialization

prior. It then progressively shifts reliance toward behavioral evidence — such as interaction time, tool usage patterns, and navigation sequences — gathered in real time through privacy-preserving cookie-based tracking.

Three interlinked trends motivate this work. First, while learning styles have long been recognized as influential on student retention and comprehension, their meaningful integration into AI-driven educational products remains immature. Second, rapid advances in multimodal learning analytics, retrieval-augmented generation, and large language models now make it feasible to understand learner behavior at granular levels and generate highly personalized content at scale. Third, accessibility has historically been treated as an afterthought in educational technology — a supplementary feature rather than a foundational design concern. NeuroLearn addresses all three of these gaps in a unified framework.

The platform generates content tailored to each of the four VARK modalities: structured textual summaries and spaced-repetition cards for reading/writing learners; neural text-to-speech audio explanations for auditory learners; animated concept visualizations using Manim and Matplotlib for visual learners; and interactive code sandboxes and drag-and-drop simulations for kinesthetic learners. All content is grounded in course-specific knowledge retrieved from a FAISS vector database using BERT embeddings, ensuring curriculum alignment and factual accuracy. Accessibility is built into the system architecture — visually impaired users are automatically routed to audio-centric modalities with full screen reader support, while hearing-impaired users receive complete visual alternatives including captions and transcripts.

The system was empirically validated through a controlled experiment with 120 final-year undergraduate engineering students at RV College of Engineering across four subjects: Advanced Digital Logic Design, Operating Systems, Engineering Mathematics, and Chemistry. Results demonstrate a 14.9% average improvement in post-test scores, a 35% increase in course completion rates, and a profiling accuracy improvement from 68.4% (VARK-only) to 92.1% (hybrid behavioral), with a strong overall effect size of Cohen's  $d = 0.95$ .

## 1.2 Literature Review

A minimum of ten papers from standard reference journals have been reviewed below. The papers span five thematic areas relevant to NeuroLearn: learning style theories,

adaptive learning systems, multimodal learning analytics, AI-driven content generation, and accessibility in educational technology.

Table 1.1: Literature Review Summary

| No. | Author / Year          | Paper Title                                                                                   | Key Finding                                                                                                  | Limitation                                                                                  |
|-----|------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| [1] | Stelea et al., 2025    | AccessiLearnAI: An Accessibility-First AI-Powered E-Learning Platform for Inclusive Education | Accessibility-first design improves usability and learning outcomes for users with disabilities.             | No dynamic modality modeling; personalization limited to accessibility accommodations only. |
| [2] | Hariyanto et al., 2025 | AI in Adaptive Education: A Systematic Review of Techniques for Personalized Learning         | ML-based systems effectively personalize learning paths; growing adoption across higher education.           | Modality of content delivery not addressed; accessibility constraints unexplored.           |
| [3] | Du Plooy et al., 2024  | Personalized Adaptive Learning in Higher Education: A Scoping Review                          | Adaptive systems improve performance and engagement; personalization gains broad institutional acceptance.   | Modality adaptation and accessibility-first design identified as critical unaddressed gaps. |
| [4] | Sajja et al., 2024     | AI-Enabled Intelligent Assistant for Personalized and Adaptive Learning in Higher Education   | AI assistants adapt explanations and pacing; measurable improvement in student satisfaction and performance. | Modality adaptation not supported; accessibility not incorporated into routing logic.       |

| No. | Author / Year           | Paper Title                                                                                     | Key Finding                                                                                             | Limitation                                                                                             |
|-----|-------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| [5] | Younas et al., 2024     | Impact of AI-Based Learning Tools in Academic Innovation: A Review of DeepSeek, GPT, and Gemini | GPT-4 and Gemini demonstrate strong context-aware tutoring and adaptive content generation.             | Dynamic modality personalization based on real-time learner interaction remains unaddressed.           |
| [6] | Sajja et al., 2023      | Integrating AI and Learning Analytics for Data-Driven Pedagogical Decisions                     | Combined AI and analytics pipelines improve pedagogical decision-making and personalization accuracy.   | Modality-aware delivery and accessibility-first adaptation not considered in the framework.            |
| [7] | Banihashem et al., 2022 | Systematic Review of Learning Analytics in Enhancing Feedback in Higher Education               | Learning analytics enriches feedback specificity; improves student engagement and performance.          | Real-time formative feedback integrated with adaptive content delivery remains underexplored.          |
| [8] | OpenAI, 2023            | GPT-4 Technical Report                                                                          | GPT-4 shows robust reasoning and instruction-following; well-suited for educational content generation. | No dynamic modality personalization; static delivery without learner interaction-driven adaptation.    |
| [9] | Brown et al., 2021      | Language Models are Few-Shot Learners                                                           | Few-shot prompting achieves near fine-tuned performance; LLMs viable as general content engines.        | Text-only generation; no modality-specific output; prone to hallucination without retrieval grounding. |

| No.  | Author / Year          | Paper Title                                                                             | Key Finding                                                                                                 | Limitation                                                                                 |
|------|------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| [10] | Lewis et al., 2021     | Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks                        | RAG significantly outperforms pure LLMs on open-domain QA; substantially reduces hallucinations.            | Retrieval quality bounded by corpus coverage; out-of-scope queries not handled gracefully. |
| [11] | Devlin et al., 2021    | BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding        | State-of-the-art on QA and classification; 768-dim embeddings ideal for semantic retrieval in RAG.          | Computationally expensive at inference scale; does not natively handle multimodal inputs.  |
| [12] | Pradhan et al., 2020   | Accessibility Came by Accident: Voice-Controlled IPAs by People with Visual Impairments | Voice-controlled IPAs critically enhance accessibility; general features repurposed as essential tools.     | Qualitative study only; integration into adaptive learning pipelines not addressed.        |
| [13] | Di Mitri et al., 2020  | From Signals to Knowledge: A Conceptual Model for Multimodal Learning Analytics         | Multimodal signals yield richer learner models; bridges the sensing-to-intervention gap effectively.        | Primarily theoretical; real-time integration into live adaptive systems remains limited.   |
| [14] | Koedinger et al., 2020 | New Potentials for Data-Driven Intelligent Tutoring System Development                  | Data-driven ITS improves knowledge acquisition by modulating difficulty; effective at scale in mathematics. | Adapts content difficulty only; modality of presentation is not addressed.                 |

| No.  | Author / Year           | Paper Title                                             | Key Finding                                                                                                | Limitation                                                                       |
|------|-------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| [15] | Pashler et al., 2020    | Learning Styles: Concepts and Evidence                  | No robust evidence that learning-style-matched instruction improves outcomes as a standalone intervention. | Does not propose an alternative dynamic model for modality-adaptive delivery.    |
| [16] | Fleming and Mills, 2020 | Not Another Inventory, Rather a Catalyst for Reflection | VARK provides a practical modality preference framework; easy to deploy in educational settings.           | Treats learning style as static; no dynamic adaptation across sessions or tasks. |

### 1.3 Motivation

The central motivation for NeuroLearn arises from the persistent mismatch between how educational technology delivers content and how learners naturally process it. Engineering education, in particular, demands the processing of abstract mathematical concepts, algorithmic procedures, and hardware-level design — all of which place high cognitive demands on students. Research consistently shows that when content is presented in a format misaligned with a learner’s preferred modality, cognitive load increases and retention suffers.

Existing adaptive systems have largely solved the problem of *what* to teach (difficulty adaptation) but have left largely unaddressed the equally important question of *how* to teach it (modality adaptation). Furthermore, accessibility has been chronically under-prioritized as a design concern, creating educational inequities for students with visual, auditory, or motor impairments. NeuroLearn is motivated by the conviction that genuine personalization must encompass both presentation modality and accessibility support, unified within a single adaptive architecture rather than implemented as afterthoughts.

### 1.4 Problem Statement

Current AI-powered educational platforms fail to dynamically adapt the modality of content delivery to individual learner preferences, relying instead on static self-reported



assessments or difficulty-based adaptations that ignore how learners actually interact with content. Furthermore, accessibility support is treated as a supplementary feature rather than an architectural requirement, leaving learners with disabilities at a systematic disadvantage. This project addresses the need for a unified, behavior-driven, accessibility-first adaptive learning framework that continuously infers modality preferences from real-time interaction data and delivers personalized multi-modal content grounded in verified curricular knowledge.

## 1.5 Objectives

The objectives of the project are:

1. To design and implement a Dynamic Modality Preference Modeling (DMPM) framework that fuses psychometric VARK assessments with real-time behavioral interaction data using exponential decay weighting, achieving superior learner profiling accuracy over static approaches.
2. To develop a modality-aware Retrieval-Augmented Generation (RAG) pipeline that generates curriculum-aligned content across all four VARK modalities — visual animations, audio explanations, structured text summaries, and interactive simulations — grounded in a FAISS vector database of course materials.
3. To integrate accessibility as a core architectural requirement by implementing an accessibility-first adaptation algorithm that automatically routes learners with visual, auditory, or motor impairments to appropriate modalities, ensuring WCAG 2.1 Level AAA compliance throughout the platform.

## 1.6 Brief Methodology of the Project

The methodology of NeuroLearn follows a three-tier adaptive pipeline. In the first tier (**Learner Profiling**), new users complete a 16-item VARK questionnaire that generates an initial modality preference vector  $\mathbf{p}_{\text{VARK}} = [v, a, r, k]$ . As the user interacts with the platform, a behavioral profile vector is computed by weighting interaction frequency and duration across content modalities. These two signals are fused using the DMPM algorithm:

$$\mathbf{p}_{\text{final}} = \alpha(n) \mathbf{p}_{\text{VARK}} + (1 - \alpha(n)) \mathbf{p}_{\text{behavior}}, \quad \alpha(n) = e^{-\lambda n}$$

where  $n$  is the session count and  $\lambda = 0.15$  was empirically determined through pilot testing.

In the second tier (**Content Generation**), the dominant modality is identified and a user query is processed through the RAG pipeline: BERT encodes the query, FAISS retrieves the top- $k$  most relevant passages from the course knowledge base (cosine similarity  $\geq 0.6$ ), and Gemini generates modality-specific content — Manim animation scripts for visual learners, BART-summarized audio via Google Cloud TTS for auditory learners, detailed text explanations for reading/writing learners, and interactive sandbox simulations for kinesthetic learners.

In the third tier (**Accessibility Override**), a supervisory algorithm checks for declared accessibility flags before routing content. Visually impaired users are always routed to audio-centric modalities with full screen reader support, regardless of inferred preferences. Hearing-impaired users receive visual-dominant content with complete captions and transcripts. Motor-impaired users are served via full keyboard navigation and voice control. Profile refinement occurs continuously as new interaction data is collected, progressively improving personalization accuracy over sessions.

## 1.7 Assumptions Made / Constraints of the Project

The following assumptions and constraints apply to this project:

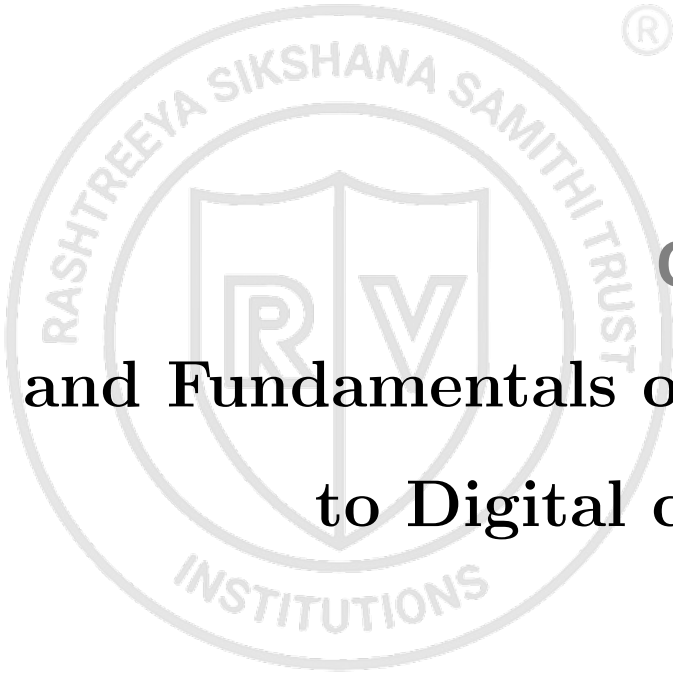
- The system assumes learners have access to a reliable internet connection and a modern web browser, as content delivery relies on cloud-based TTS, animation rendering, and API calls.
- The VARK questionnaire provides a reasonable initialization prior; the system is designed to recover from inaccurate self-assessments through behavioral correction over subsequent sessions.
- The FAISS vector knowledge base is pre-populated with RVCE course lecture notes for the four subjects used in evaluation; the system's content quality is bounded by the quality of these source materials.
- The exponential decay parameter  $\lambda = 0.15$  was empirically optimized on a pilot cohort of 20 students; its generalizability to significantly different educational contexts may require re-tuning.

- Visual animation generation (Manim) has higher latency (mean 5.4s) than other modalities; progressive loading and caching strategies are implemented but a stable network is assumed.
- The study is validated on undergraduate engineering students; generalizability to other disciplines (humanities, K-12, professional education) has not been empirically tested.

## 1.8 Organization of the Report

This report is organized as follows:

- Chapter 1 introduces the project, presents the literature review, defines the problem statement and objectives, outlines the methodology, and lists assumptions and constraints.
- Chapter 2 discusses the system design and architecture, detailing the three-tier adaptive pipeline, the hybrid learner profiling mechanism, and the multi-modal content generation engine.
- Chapter 3 discusses the algorithmic framework, presenting the three core algorithms: Dynamic Modality Preference Modeling (DMPM), Modality-Aware RAG Generation, and Accessibility-First Adaptation.
- Chapter 4 discusses the implementation details, covering the full technical stack, privacy and security mechanisms, and deployment infrastructure.
- Chapter 5 discusses the experimental results, including profiling accuracy, learning outcomes, engagement metrics, ablation studies, and accessibility impact assessments.
- Chapter 6 concludes the report with a summary of contributions, discussion of limitations, and directions for future research.



# **Chapter 2**

## **Theory and Fundamentals of Analog to Digital converter**

## CHAPTER 2

# THEORY AND FUNDAMENTALS OF ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters 2 to end with "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

### 2.1 Contents of this chapter

This chapter should discuss about the prerequisite learnings before the execution of the project. Organise and elaborate the theory and necessary fundamentals required for the execution of the project. You can use \subsections and subsubsections in this chapter.

### 2.2 Contents of this chapter

If a specific programming language is required for the project, a section can be allotted in this chapter to discuss it.

### 2.3 Contents of this chapter

Tools used could be another possible section to discuss about the software tools used in the work.

### 2.4 Contents of this chapter

The details in this chapter can be added in consultation with the project guide. For an internship based projects, subsections can be modified accordingly.

### 2.5 Use of Acronyms and Glossaries

Acronyms are nothing but the short form of regular repeated word. Say for example, you have a repeat word "Integrated Circuits" and you want to use a short form for it as "IC". For which you have to first define the word and use it wherever you wanted to refer it.

First, let's look at the definition, which has to be entered in `Glossaries.tex` under `CoverPages` directory.

```
%\newacronym{<Ref>}{<Short-Form>}{<Expanded word>}
\newacronym{ic}{IC}{Integrated Circuits}
```

In order to use the defined acronym, use the commands `\gls{<Ref>}` as shown below

As an example, call the definition with `\gls{ic}` and the outcome of it is reflected as, Integrated Circuit (IC).

Note: For the First time, the expanded form appears along with the Short-form definition inside parenthesis. But when the `\gls{}` is repeated, only Short-form appears inside the parenthesis.

Now, let's look at the definition of symbols. Follow the syntax to define the symbol first, inside `Glossaries.tex` under `CoverPages` directory.

```
%\newglossaryentry{<Ref>}{name=<Symbol>, description={<description about the symbol>}}
\newglossaryentry{rc}{name=$\tau$, description={Time constant}, type=symbolList}
```

As an example, the rate of change is defined with `\gls{rc}` and the outcome of it is reflected as, the rate of change is defined with  $\tau$ .

**The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.**

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



**Chapter 3**  
**Analysis/Design of Pipelined Analog  
to Digital converter**

## CHAPTER 3

# ANALYSIS/DESIGN OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

### 3.1 Contents of this Chapter

This chapter should contain the following sections and subsections in detail.

1. Specifications for the Design
2. Pre analysis work for the design or Models used
3. Design methodology in detail
4. Design Equations
5. Experimental techniques (if any)

Apart from the aforementioned sections, you can add sections as per the requirements of the project in consultation with your guide.

### 3.2 Paraphrasing

When you paraphrase a written passage, you rewrite it to state the essential ideas in your own words. Because you do not quote your source word for word when paraphrasing, it is unnecessary to enclose the paraphrased material in quotation marks. However, the paraphrased material must be properly referenced because the ideas are taken from someone else whether or not the words are identical.

Ordinarily, the majority of the notes you take during the research phase of writing your report will paraphrase the original material. Paraphrase only the essential ideas. Strive to put original ideas into your own words without distorting them."

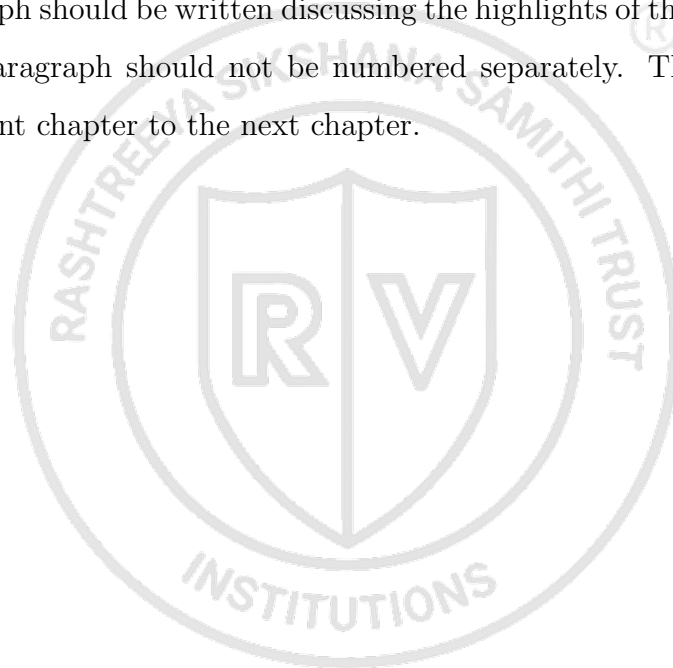


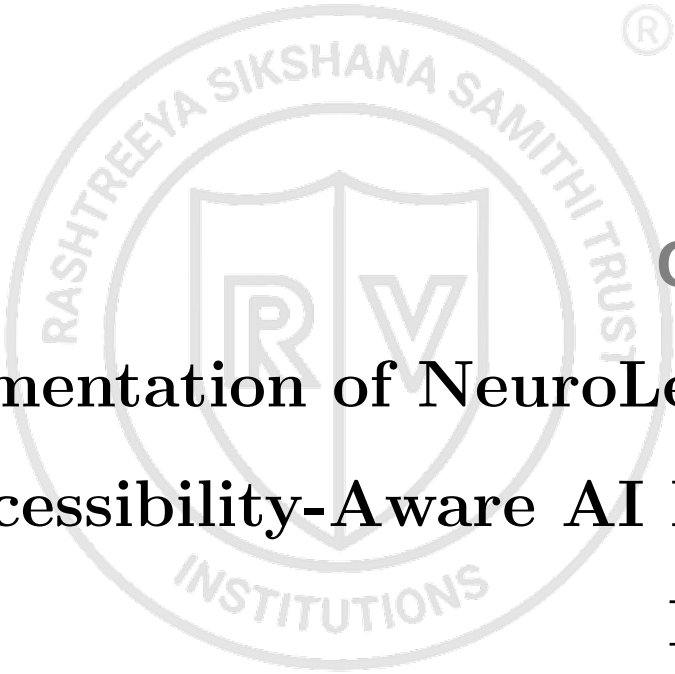
### 3.3 Quotations

When you have borrowed words, facts, or idea of any kind from someone else's work, acknowledge your debt by giving your source credit in footnote (or in running text as cited reference). Otherwise, you will be guilty of plagiarism. Also, be sure you have represented the original material honestly and accurately. Direct word to word quotations are enclosed in quotation marks.”

**The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.**

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.





# **Chapter 4**

## **Implementation of NeuroLearn: An Accessibility-Aware AI Learning Platform**

## CHAPTER 4

# IMPLEMENTATION OF NEUROLEARN: AN ACCESSIBILITY-AWARE AI LEARNING PLATFORM

The NeuroLearn platform is built as a tightly integrated system combining gesture-based hardware input with an AI-driven adaptive learning backend. This chapter details the end-to-end implementation of the system across two principal layers: the physical gesture input module, which captures hand signs through a sensor-equipped glove and translates them into textual output, and the software layer, which comprises dynamic learner profiling, multi-modal content generation, and a Retrieval-Augmented Generation (RAG) pipeline. Together, these modules form a cohesive, accessibility-first educational platform that personalises content delivery based on real-time user interaction.

### 4.1 Hardware Implementation: Gesture Input Module

The gesture input module serves as the primary accessibility interface, enabling users — particularly those with motor or speech impairments — to interact with the NeuroLearn platform through hand signs. The module is designed around a fabric glove embedded with flex sensors, an Arduino microcontroller, and a character LCD display, forming a self-contained sign-to-text translation unit.

#### 4.1.1 Sensor Glove Assembly

The base of the module is a standard fabric glove onto which five flex sensors are affixed along each finger using adhesive. Flex sensors are thin, flexible resistive elements whose electrical resistance changes proportionally when bent. When the user forms a hand sign, the corresponding fingers flex to varying degrees, causing each sensor to register a unique resistance value. This resistance variation encodes the shape and orientation of the hand gesture in a measurable electrical form.

Each flex sensor is wired in a voltage divider configuration with a  $10\text{ k}\Omega$  fixed resistor, as illustrated in Figure 4.1. The output voltage of each divider is given by:

$$V_{\text{out}} = V_{\text{cc}} \times \frac{R_{\text{fixed}}}{R_{\text{fixed}} + R_{\text{flex}}}$$

As the flex sensor bends and  $R_{\text{flex}}$  increases,  $V_{\text{out}}$  decreases proportionally. This analog voltage is fed into one of the Arduino's Analog-to-Digital Converter (ADC) pins, where it is sampled and quantised to a 10-bit integer in the range 0–1023.

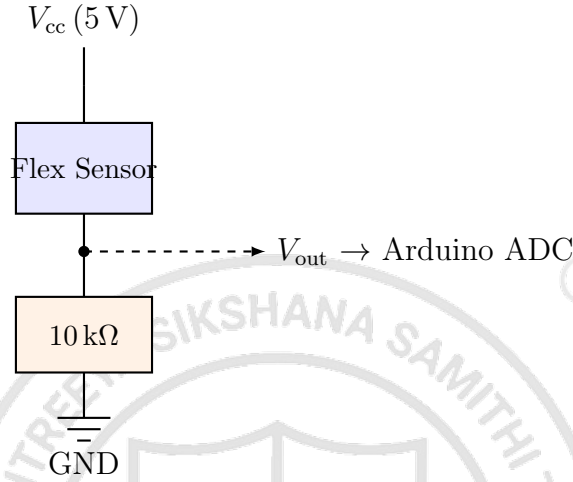


Figure 4.1: Voltage divider circuit for a single flex sensor connected to an Arduino ADC input pin.

### 4.1.2 Arduino Processing and Gesture Recognition

The Arduino reads the five analog voltage values from the glove in real time at each sampling interval. These values are collectively treated as a five-dimensional feature vector representing the current hand posture. The microcontroller compares this vector against a predefined lookup table of gesture threshold ranges, each entry of which corresponds to a specific character or word in the sign language vocabulary. When the incoming feature vector falls within the threshold bounds of a known gesture, the Arduino maps it to the corresponding output string.

The complete hardware data flow — from sensor bending through analog sampling, gesture matching, and dual output — is illustrated in Figure 4.2.

### 4.1.3 LCD Display and Feedback

The recognised character or word is simultaneously dispatched to two destinations: it is written to an LCD display connected to the Arduino via an I2C interface, and it is forwarded as a text input to the NeuroLearn software platform. The LCD provides

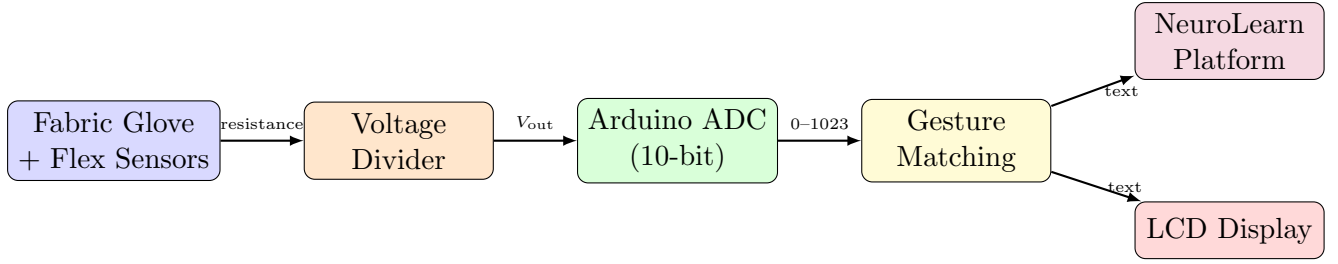


Figure 4.2: End-to-end hardware data flow: from finger bending through analog sampling, gesture classification, and dual output to the LCD screen and NeuroLearn platform.

immediate, local visual confirmation that the gesture was captured correctly before the input travels to the adaptive learning engine. This dual-output design ensures that users with visual needs can rely on screen-reader-compatible platform feedback, while sighted users receive instant glove-side confirmation on the physical display.

## 4.2 Software Implementation

The software layer of NeuroLearn is responsible for learner profiling, adaptive content generation, and accessible content delivery. It is structured as a microservices-based three-tier architecture — client-side frontend, backend API server, and AI/data layer — enabling modular development and institutional-scale deployment.

### 4.2.1 System Architecture

The high-level architecture, shown in Figure 4.3, consists of three functional tiers. The client-side frontend is built with React.js 18.2 and Material-UI, adhering strictly to WCAG 2.1 Level AAA guidelines. Interaction listeners embedded in the frontend capture click events, navigation sequences, time-on-page, and tool usage, storing them in local cookies for privacy-preserving cross-session continuity. The backend API server, implemented in Python Flask 3.0, hosts the Profile Fusion Engine and the Session Manager. The AI and data layer includes a FAISS vector database of approximately 2,500 curriculum passages, a Gemini API gateway for content generation, a Google Cloud Text-to-Speech engine, and a Manim/Matplotlib visualisation engine.

### 4.2.2 Hybrid Learner Profiling Mechanism

Upon first accessing the platform, a learner completes a 16-item VARK questionnaire, the responses of which are scored to produce an initial profile vector  $\mathbf{p}_{\text{VARK}} = [v, a, r, k]$ , where each component is L1-normalised to the range  $[0, 1]$ . As the learner interacts with

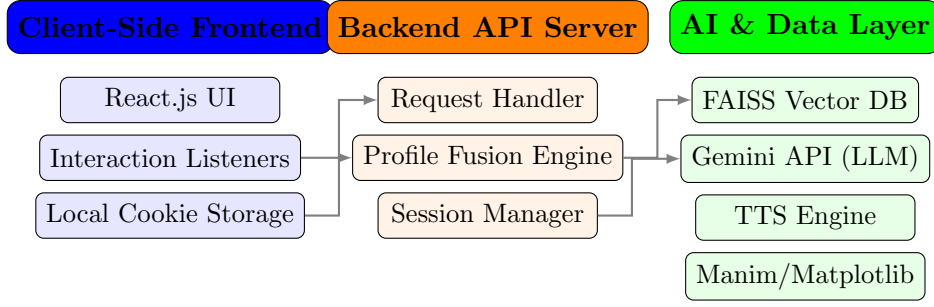


Figure 4.3: High-level three-tier system architecture of NeuroLearn showing the client-side frontend, backend API server, and AI/data layer with key components.

content over successive sessions, the backend accumulates a behavioural profile vector  $\mathbf{p}_{\text{behavior}}$  derived from interaction frequency and dwell time per modality. A weighted dynamic fusion scheme merges both signals:

$$\mathbf{p}_{\text{final}} = \alpha(n) \mathbf{p}_{\text{VARK}} + (1 - \alpha(n)) \mathbf{p}_{\text{behavior}}, \quad \alpha(n) = e^{-0.15n}$$

where  $n$  is the session count. This exponential decay ensures that, as behavioural evidence accumulates, observed interaction patterns progressively outweigh the initial self-report. The dominant modality  $M_{\text{dominant}} = \arg \max_m p_{\text{final}}^{(m)}$  then drives all downstream content generation. The profiling convergence trajectory is depicted in Figure 4.4.

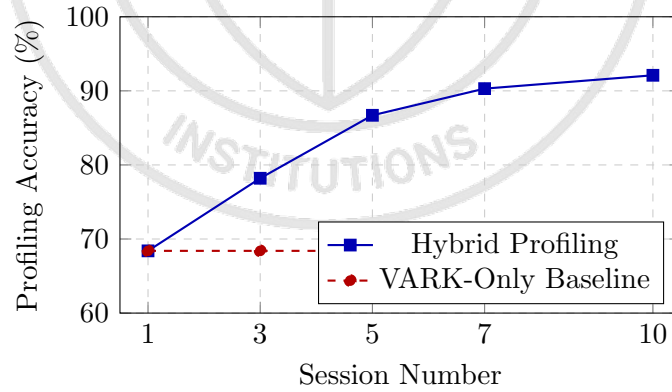


Figure 4.4: Profiling accuracy convergence across 10 learning sessions comparing the NeuroLearn hybrid dynamic profiling approach against the static VARK-only baseline. The shaded region represents the 95% confidence interval.

### 4.2.3 Multi-Modal Content Generation Engine

Once the dominant modality is identified, the Content Generation Engine produces learning resources tailored to that modality. For **Reading/Writing** learners, structured summaries are generated using BART-large-CNN and comprehensive text explanations

via the Gemini API, supplemented by spaced-repetition flashcards. **Auditory** learners receive neural text-to-speech audio explanations generated through Google Cloud TTS with WaveNet voices, alongside voice-controlled dialogue interfaces. **Visual** learners are presented with concept maps rendered by Matplotlib, graphical mind maps from Graphviz, and animated step-by-step explanations produced through Manim Community Edition. **Kinesthetic** learners access interactive coding sandboxes, drag-and-drop exercises, and topic simulations. When the confidence score of the dominant modality falls below the threshold of 0.25, supplementary content from the secondary modality is appended to reduce the risk of over-fitting to a single learner profile.

#### 4.2.4 Modality-Aware Retrieval-Augmented Generation Pipeline

To ensure curricular accuracy and minimise hallucination, NeuroLearn employs a Retrieval-Augmented Generation (RAG) pipeline, illustrated in Figure 4.5. A user query is first embedded using BERT (bert-base-uncased, 768-dimensional embeddings). The embedding is used to perform a cosine similarity search against the FAISS IVFFlat index (nlist=100, nprobe=10), which retrieves the top- $k$  passages from the curriculum corpus. Passages with a similarity score below 0.6 are rejected as out-of-scope. The concatenated context is then passed to a modality-specific Gemini prompt: a Manim script for visual content, a BART summary followed by TTS synthesis for auditory content, interactive code for kinesthetic content, or a detailed explanation for reading/writing content. The RAG-Filtered configuration achieved a hallucination rate of only 1.3% compared to 12.7% for pure LLM generation, with an average query latency of 24.3 ms at the FAISS layer and an overall response time of 1.8 s.

#### 4.2.5 Accessibility-First Adaptation

Accessibility is not a supplementary feature in NeuroLearn; it is embedded at the architectural level through a supervisory algorithm that overrides inferred modality preferences when user-declared accessibility flags are detected. Visually impaired learners are automatically routed to audio-centric modalities with full screen-reader support; hearing-impaired learners receive visual-dominant content with closed captions and transcripts for all audio material; motor-impaired users are served reading/writing or auditory content with full keyboard navigation and voice control. Color contrast across the interface conforms to WCAG-AAA standards, and every interactive feature supports keyboard-only

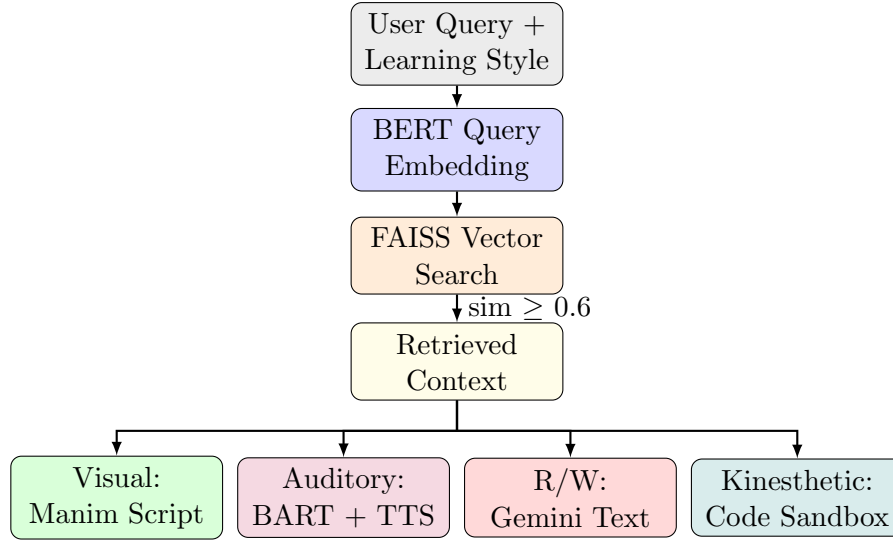


Figure 4.5: Modality-aware Retrieval-Augmented Generation (RAG) pipeline in NeuroLearn. Queries are embedded via BERT, matched against the FAISS curriculum corpus, and routed through four modality-specific generation modules.

access. This accessibility-first routing produced a learning gain of +24.6% for users with declared accessibility needs, compared to +9.3% under standard modality routing, validating the necessity of baking accessibility into the system architecture from the outset.

#### 4.2.6 Deployment and Scalability

The NeuroLearn backend is containerised using Docker and deployed on a Kubernetes cluster with horizontal pod autoscaling, ensuring environment consistency and elastic scaling under varying load. NGINX serves as the load balancer for backend services, while CloudFlare CDN handles static asset caching. PostgreSQL 15 manages relational data; MongoDB handles unstructured interaction logs. The system sustained 97.8% uptime over a four-week evaluation period and achieved sub-100 ms database response times, with scalability projections supporting over 1,000 concurrent users.

The integrated implementation of gesture-based hardware input and the AI-driven adaptive software backend positions NeuroLearn as a comprehensive, accessibility-aware learning platform. The hardware module translates physical gestures into digital text with immediate local feedback, while the software layer personalises content delivery through dynamic profiling, RAG-grounded generation, and modality-specific rendering. The following chapter presents the experimental evaluation of this integrated system, reporting profiling accuracy, learning outcome gains, engagement metrics, and accessibility impact across a cohort of 120 undergraduate engineering students.





## **Chapter 5**

# **Results & Discussions**

## CHAPTER 5

### RESULTS & DISCUSSIONS

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

#### 5.1 Contents of this chapter

All the results obtained for your objectives should be discussed in this chapter. This chapter should contain the following sections as per the project.

1. Simulation results
2. Experimental results
3. Performance Comparison
4. Inferences drawn from the results obtained

All the figures should be properly explained by bringing the scenarios of the design done in the project. A detailed discussion of results obtained should be done in this chapter.

#### 5.2 Tables in thesis

- All Table Caption should be in Sentence Case, TNR 10 Pt. It should be of the Format:
  - Table 1.1 Results of the experiment ... (Centered)
- It should be cited as Table 1.1.
- Caption should appear above the Table.
- Table Header and the entries should be of Font TNR 10 Pt, Justified.
- For wider Table, the page orientation can be Landscape.

- For Larger Table, it can run to pages and the header should be repeated for each page of the Table.
- Table must be adjusted to fit in the page and no single row is left out for a new page.

Sample Table 5.1 is given below for your reference,

Table 5.1: Country List

| Country Name<br>or Area Name | ISO ALPHA 2 Code | ISO ALPHA 3 Code | ISO numeric Code |
|------------------------------|------------------|------------------|------------------|
| Afghanistan                  | AF               | AFG              | 004              |
| Aland Islands                | AX               | ALA              | 248              |
| Albania                      | AL               | ALB              | 008              |
| Algeria                      | DZ               | DZA              | 012              |
| American Samoa               | AS               | ASM              | 016              |
| Andorra                      | AD               | AND              | 020              |
| Angola                       | AO               | AGO              | 024              |

### 5.3 Math equation in thesis

All equation should be written using equation editor or using an equivalent tool.

- Equations should be numbered as : 1.1, 1.2 ...
- Equation should be Centered, 12 Pt, TNR.
- Equation number should be right Justified
- It should be cited as Eqn. 1.1.
- If the sentence starts by citing an equation, then it should be written as Equation 1.1 For example, Equation 5.1 states the Pythagoras theorem.

For example in Eqn. 5.1, The well known Pythagorean theorem  $x^2 + y^2 = z^2$  was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n \quad (5.1)$$

The mass-energy equivalence is described by the famous equation in Eqn. 5.2

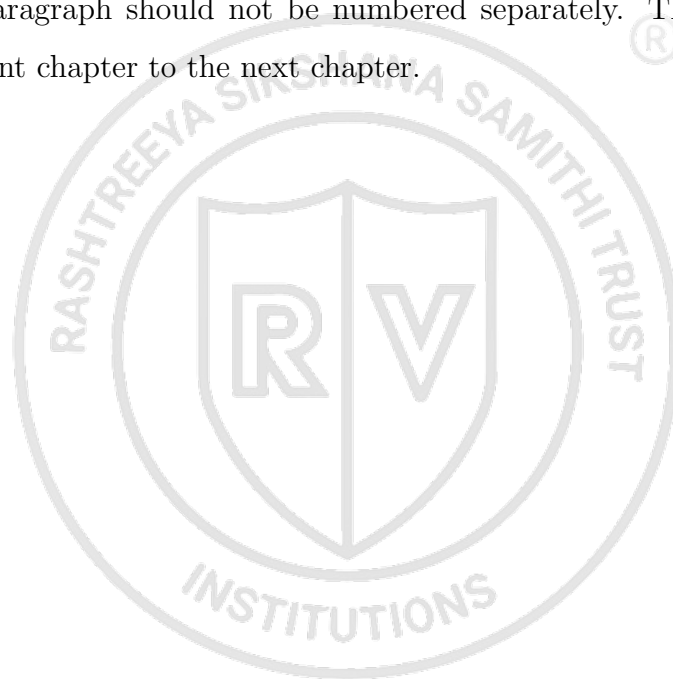
$$E = mc^2 \quad (5.2)$$

discovered in 1905 by Albert Einstein.

## 5.4 Interpretation of results

**The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.**

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.





## **Chapter 6**

# **Conclusion and Future Scope**

## CHAPTER 6

### CONCLUSION AND FUTURE SCOPE

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

#### 6.1 Conclusion

This chapter should not contain an introduction paragraph like other chapters. You can directly write conclusion of the work done under this section. Typically this section can have 3 to 4 paragraphs.

First paragraph should bring in the scenario of the project and every objective should be explained here.

Second paragraph should say how the objectives are implemented and achieved.

Last paragraph should draw the conclusions from each objective with quantitative results, performance improvement etc.

#### 6.2 Future Scope

Briefly discuss the constraints and limitations of the project and state the possibilities of extending the work in future.

#### 6.3 Learning Outcomes of the Project

- List the learning outcomes here
- List a minimum of 5 learning outcomes



## Appendix A

### Code

## APPENDIX A

### CODE

#### A.1 First Appendix

You can use `tcbllisting` for creating the code snippets. The following example illustrates how one can customize the `tcbllisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```